LI/LH Multivariable & Vector Analysis (2MVA(3))

Problems for week 7

(Questions on this problem sheet are NOT assessed)

Problem 1. True or false? If $\boldsymbol{A}$ and $\boldsymbol{B}$ are vectors in 3-dimensional space, then $\boldsymbol{A}\times(\boldsymbol{A}\times\boldsymbol{B})$ is always zero.

Problem 2. Consider the triangle in 3-dimensional space with vertices $P = (1, 1, 1)$, $Q = (2, 2, 2)$, $R = (3, 1, 2)$. What is its area?

Problem 3. Given the points $P = (1, 1, -1)$, $Q = (1, 2, 0)$ and $R = (-2, 2, 2)$, find:

(a) $\overrightarrow{PQ} \times \overrightarrow{PR}$;

(b) a plane $ax + by + cz = d$ through the points P, Q, R.

Problem 4. Given vectors $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ in 3-dimensional space, show that

$$\det(\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}) = \boldsymbol{A} \cdot (\boldsymbol{B} \times \boldsymbol{C}).$$

Problem 5. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be any vectors. Prove:

(a) Cauchy–Schwarz inequality:

$$|\boldsymbol{A} \cdot \boldsymbol{B}| \leq |\boldsymbol{A}||\boldsymbol{B}|.$$

(b) Triangle inequality:

$$|\boldsymbol{A} + \boldsymbol{B}| \leq |\boldsymbol{A}| + |\boldsymbol{B}|.$$

Problem 6. Find a parametric representation of a curve C defined as

$$x^3 + 2x^2 + y^2 = 3, \qquad y \geq 0,$$

i.e., describe it in the form $x = x(t)$, $y = y(t)$. Find the relevant range of t.

Problem 7. Suppose that the position of one particle at time t is given by

$$x_1 = 3\sin t \quad y_1 = 2\cos t \quad 0 \le t \le 2\pi,$$

and the position of a second particle is given by

$$x_2 = -3 + \cos t \quad y_2 = 1 + \sin t \quad 0 \le t \le 2\pi.$$

(a) Graph the paths of both particles. How many points of intersection are there?
(b) Are any of these points of intersection *collision points*? In other words, are the particles ever at the same place at the same time? If so, find the collision points.
(c) Describe what happens if the path of the second particle is given by

$$x_2 = 3 + \cos t \quad y_2 = 1 + \sin t \quad 0 \le t \le 2\pi.$$